

Pathogen Genomics Training: Bacterial Genomics & Outbreak Investigation



Bacterial Genomics & Outbreak Investigation Installation Guide

INTRODUCTION

Many bioinformatics tools are developed primarily for Linux. In this course, we will run the tools in a Unix-like environment and install software using **Miniforge**, a lightweight Conda distribution that defaults to the **conda-forge** channel (community-maintained packages with fewer licensing restrictions). This helps ensure participants can reproduce the same software environment across different machines.

Windows users will use **Ubuntu on Windows Subsystem for Linux (WSL)**. **Mac users** can install most tools directly on macOS via Conda/Miniforge, but a few tools may behave differently or may not be available for macOS. If that happens, the course instructor will provide an alternative method.

To learn more about Miniforge and Conda, see:

<https://www.fabriziomusacchio.com/blog/2025-07-03-miniforge/> and <https://docs.conda.io/projects/conda/en/latest/user-guide/index.html>

MINIMUM SYSTEM REQUIREMENTS

- **RAM:** 8 GB minimum; 16 GB or more recommended
- **Free storage:** at least 100–200 GB minimum; 500 GB recommended (internal or external drive)
- **Operating system:** Linux, macOS, or Windows with WSL2



8 GB RAM should be sufficient for the workshop dataset. However, 16 GB or more is recommended for smoother performance.

The host-filtering step in module 1 uses a reduced human reference index to keep memory requirements manageable. Workflows using the full human reference genome, larger datasets, or many samples running in parallel may require more RAM and storage.

▼ UBUNTU INSTALLATION FOR WINDOWS USERS

- Open Windows Powershell in administrator mode. (Use Windows search and type "powershell")
- To install WSL, run the command:

```
wsl --install
```

- This installs Ubuntu on WSL2 (recommended)
- When prompted, create a new Ubuntu username and password (you won't see the password as you type).



Please write this down

- Once installation is completed, reboot the machine.

▼ MINIFORGE INSTALLATION ON UBUNTU FOR WINDOWS USERS

- Open Ubuntu terminal from Windows Start menu. Click on the “Windows” button and search for “ubuntu”. Clicking on the application will open an Ubuntu terminal, which is a LINUX terminal.
- If prompted, enter your username and password from step 1.
- Download the latest Miniforge installer for Linux (for more information, see: <https://github.com/conda-forge/miniforge>)

```
wget https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-Linux-x86_64.sh
```

- Run the installer:

```
bash Miniforge3-Linux-x86_64.sh
```



Follow the prompts to complete the installation.

Type “yes” when asked to initialize Conda.

This ensures that conda is available in new terminals.

- Close and reopen the terminal to apply changes
- If you have followed the instructions properly, you should see **(base)** at the beginning of the command line. This means that Miniforge is already activated.
- Otherwise, activate conda:

```
conda activate
```

- Important! Check your channels:

```
conda config --show channels
```

- You should only see:

- conda-forge

▼ MINIFORGE INSTALLATION FOR MAC USERS

- Open **Terminal** (Applications > Utilities > Terminal).
- Check which Mac you are using (Intel vs Apple Silicon):

```
uname -m
```

- Download the latest **Miniforge** installer for macOS (for more information, see: <https://github.com/conda-forge/miniforge>) and run the installer
 - **Apple Silicon (M1/M2/M3):**

```
curl -L -O https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-MacOSX-arm64.sh
```

```
bash Miniforge3-MacOSX-arm64.sh
```

- **Intel Mac:**

```
curl -L -O https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-MacOSX-x86_64.sh
```

```
bash Miniforge3-MacOSX-x86_64.sh
```



Follow the prompts to complete the installation.

Type “yes” when asked to initialize Conda.

This ensures that conda is available in new terminals.

- Close and reopen Terminal to apply changes.
- If you have followed the instructions properly, you should see (base) at the beginning of your command line. This means that Miniforge is already activated.

- Otherwise, activate conda:

```
conda activate
```

- Important! Check your channels:

```
conda config --show channels
```

- You should only see:

```
- conda-forge
```

▼ DOWNLOAD ENVIRONMENT FILES

- Open a new terminal and check your current location. `pwd` means *print working directory*

```
pwd
```

- You may see something similar to:

```
/home/your_username
```

- This is your home directory
- Create a folder named "course"

```
mkdir course
```

- Navigate to the folder

```
cd course
```

- Install the `unzip` utility, keying in your ubuntu password when prompted.

```
sudo apt install unzip
```

- Download the environment files

```
wget https://mendel3.bii.a-star.edu.sg/METHODS/HEIDI/nepal_pgt2026/env_files.zip
```

- Extract the env_files directory

```
unzip env_files.zip
```

- This creates a new folder env_files/ in your directory. List the extracted YAML files

```
ls -lth env_files/*.yaml
```

- You should see something like this:

```
-rw-r--r-- 1 your_username your_username 372 Aug 4 07:54 e
nv_files/typhi-amr.yaml
-rw-r--r-- 1 your_username your_username 245 Aug 4 07:54 e
nv_files/typhi-phylo.yaml
-rw-r--r-- 1 your_username your_username 233 Aug 4 07:54 e
nv_files/typhi-qc.yaml
```

- Proceed with creating environments for the tutorials.

▼ CREATING ENVIRONMENTS AND INSTALLING TOOLS



For reproducibility, we use YAML environment files so that everyone installs the same tools and their dependencies across different computers. **However, for macOS (Apple) users:** When creating the Conda environments from the provided YAML files, installation may fail on some Macs. This has been observed on M1 systems because some bioinformatics dependencies do not have native osx-arm64 builds available.

If you encounter solver errors or “package not found” messages, please inform the on-site instructors — they will provide an alternative setup method.

- After installing Miniforge, your terminal should open in “base” environment.



It is not advisable to download anything on base. Instead, create a new environment for any new project or workflow.

- Navigate to env_files directory

```
cd env_files
```

- List all files in this directory:

```
ls -l
```

- You should see 3 environment files

```
typhi-qc.yml  
typhi_amr.yml  
typhi_phylo.yml
```

- These files are needed for the 3 modules in this course

Module No.	Module Name	Environment file
1	Quality Assessment and Genome Assembly	typhi-qc.yml
2	Antimicrobial Genomics	typhi-amr.yml
3	Phylogenetics and Outbreak Analysis	typhi-phylo.yml

▼ Module 1 Setup

- Create typhi-qc environment. In the command, **-n** indicates your desired environment name while **-f** indicates the shared environment file.

```
conda env create -n typhi-qc -f typhi-qc.yml
```

- Follow the on-screen instructions to download all the packages needed.
- Once completed, activate the environment for module 1.

```
conda activate typhi-qc
```

- To view the installed packages within the environment, run:

```
conda list
```

- For this module, **Bactinspector** requires an older version of pandas, a Python software library with tools to allow working with tabular data. Downgrade pandas to version **1.5.3** by installing pandas=1.5.3 individually using conda install and follow on-screen instructions:

```
conda install -c conda-forge pandas=1.5.3
```

- The table below summarizes all the programs that you have installed and their main use for module 1.

Program	Main use
FastQC	Assesses the quality of raw sequencing reads and produces reports on read quality, adapter content, GC content and other metrics.
fastp	Cleans and filters sequencing reads, including adapter removal, quality trimming and removal of low-quality reads.
Minimap2	Aligns sequencing reads or assembled genomes against a reference sequence.

Program	Main use
SAMtools	Processes sequence alignment files, especially SAM and BAM files. It can sort, index, filter and summarize alignments.
Shovill	Performs bacterial genome assembly from Illumina sequencing reads using an optimized assembly workflow.
QUAST	Evaluates the quality of genome assemblies using measures such as number of contigs, total assembly length and N50.
MultiQC	Combines results from multiple quality-control programs into a single summary report.
Mash	Rapidly compares genomes or sequencing datasets using sequence sketches. It is commonly used for species identification and similarity checking.
BactInspectorMax	Performs bacterial sequencing quality assessment and checks characteristics such as species identity, contamination and assembly quality.

▼ Module 2 Setup

- It is not necessary to deactivate your current environment before creating a new one. Conda creates the new environment separately. However, if you prefer to exit your previous environment, use:

```
conda deactivate
```

- Create typhi-amr environment. In the command, **-n** indicates your desired environment name while **-f** indicates the shared environment file.

```
conda env create -n typhi-amr -f typhi-amr.yml
```

- Follow the on-screen instructions to download all the packages needed.
- Once completed, activate the environment for module 2.

```
conda activate typhi-amr
```

- The table below summarizes the programs you have downloaded for module 2.

Program	Main use in the workflow
RGI	Detects antimicrobial-resistance genes and resistance-associated mutations using the CARD database.
AMRFinderPlus	Identifies AMR genes, point mutations, stress-response genes, virulence factors, and other resistance-related features using NCBI databases.
hAMRonization	Converts outputs from different AMR tools into one standardized PHA4GE-compatible format so results can be compared or combined.
PlasmidFinder	Detects plasmid replicon sequences and assigns likely plasmid incompatibility groups, helping assess whether AMR genes may be plasmid-associated.

Program	Main use in the workflow
Prodigal	Predicts protein-coding genes in assembled bacterial contigs. RGI uses it internally when contig input is supplied.
BLAST	Compares sequences against reference databases. It supports gene and plasmid-replicon detection in several tools.
DIAMOND	Performs fast protein-sequence similarity searches. RGI may use it as a faster alternative to protein BLAST.
HMMER	Searches for conserved protein-family patterns using hidden Markov models. AMRFinderPlus uses HMMs to detect diverse gene families.
seqkit	Command-line toolkit for inspecting and manipulating FASTA and FASTQ files
wget	Command line utilities tool for downloading database files.

▼ Module 3 Setup

- Create typhi-phylogeny environment.

```
conda env create -n typhi-phylo -f typhi-phylo.yml
```

- Follow the on-screen instructions to download all the packages needed.
- For module 3, you will need an additional program called snippy. However, this will require it's own environment due conflicts between the program's dependencies and with other programs in the typhi-phylo environment.

```
conda create -n snippy -c conda-forge -c bioconda snippy
```

- The table below summarizes the programs you have downloaded for Module 3.

Program	Main use
MAFFT	Align gene or locus sequences for multiple sequence alignment.
Snippy	Map reads to the CT18 reference, call variants, and create pseudogenomes/core alignments.
Gubbins	Detect and remove recombinant regions from the whole-genome alignment.
IQ-TREE 2	Build a maximum-likelihood phylogenetic tree and assess branch support.
snp-sites	Extract variable/SNP sites from an alignment.
MLST	Assign the broad seven-locus sequence type from assembled contigs.
Mykrobe / GenoTyphi	Assign Typhi-specific genotypes directly from FASTQ reads; also report selected AMR and plasmid markers.
snp-dists	Calculate pairwise SNP distances between isolates.

- Congatulations! You should now see all **four** environments when you run:

```
conda env list
```



Further learning:

<https://docs.conda.io/projects/conda/en/latest/user-guide/cheatsheet.html>

▼ DOWNLOAD FIGTREE

- After completing Module 3, you will also require a program to view your tree.
- While you may view your trees using a web-based viewer such as [iTOL](#) , we recommend downloading FigTree is a free desktop application for viewing and annotating phylogenetic trees (e.g., **.tree**, **.newick**, **.nwk** files).
- Download [FigTree \(from GitHub releases\)](#).
- Choose the correct installer for your operating system:
 - **Windows:** download the **.exe** installer
 - **macOS:** download the **.dmg** installer
- Install and launch FigTree.
- Open a tree file:
 - In FigTree, go to **File** → **Open...** and select your tree file.

▼ DOWNLOAD THE DATASET

